

Introduction



Theory

The term transdisciplinarity is characterized by a wide variety of nuanced definitions.

In the TRAIL, we use the definition from the OECD (2021):

"The integration of academic researchers from different disciplines with non-academic partners to jointly create new knowledge and theories in pursuit of a common goal."

This definition highlights three essential features that distinguish transdisciplinarity from other forms of collaboration:

1. Type of collaboration partners: Academic and non-academic partners learn, work, and conduct research together in transdisciplinary settings.
2. Nature of the problem addressed: The focus is on complex societal issues without clear-cut solutions, often referred to as "wicked problems."
3. Nature of involvement: Different partners learn and work on an equal footing, genuinely collaborating to generate new knowledge and theories.

This TRAIL has a special focus on transdisciplinary education.

At TRAIL, we define transdisciplinary education as:

"The integration of researchers, teachers, and students from different disciplines with non-academic participants who co-create insights, new knowledge, or products to achieve a common goal."

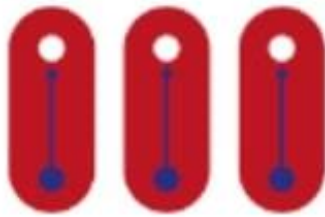
In transdisciplinary education, teachers and students are also considered collaboration partners. This differs from transdisciplinary research, which primarily involves researchers from various disciplines working with non-academic partners to create new knowledge and innovations. TRAIL aims to serve all interests equally. At the same time, the inclusion of students presents challenges in realizing this equality, due to differing interests and power dynamics, which will be further discussed in the base camp.

Inter- or transdisciplinary: common confusions

Academic literature and dialogues with experts emphasize the lack of a shared understanding of transdisciplinarity (Klein, 2018).

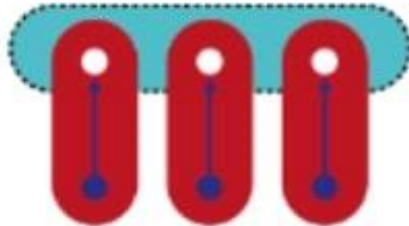
In particular, the distinction from interdisciplinary collaboration is not always clear. While both approaches aim to integrate different types of knowledge, transdisciplinarity is distinguished by the involvement of non-

academic stakeholders as equal participants in the same project. In transdisciplinarity, the diversity of perspectives and "types of knowledge" is greater, and the problems addressed are generally more complex. As complexity increases, so do the challenges and learning opportunities in collaboration (see figure below).



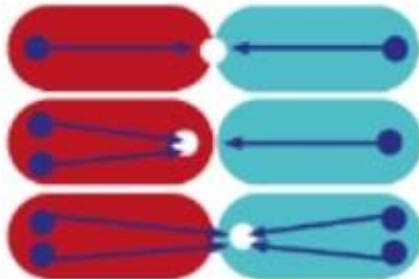
Disciplinair

- In één academische discipline
- Disciplinaire doelstelling
- Ontwikkelt nieuwe disciplinaire kennis



Multidisciplinair:

- Verschillende disciplines
- Verschillende disciplinaire doelstellingen onder een overkoepelend thema



Participatief

- Academische en non-academische deelnemers
- Kennisuitwisseling zonder integratie



Interdisciplinair

- Overstijgt disciplinaire grenzen
- Ontwikkelt geïntegreerde kennis



Transdisciplinair

- overstijgt disciplinaire en wetenschappelijke sectoriale grenzen
- gemeenschappelijke doelstelling
- Ontwikkelt geïntegreerde kennis voor wetenschap en maatschappij

● stakeholder deelnemers
● Disciplines

○ doel, gemeenschappelijke kennis
■ academische kennis ■ maatschappelijke kennis

Bron:

Morton, L. W., Eigenbrode, S. D., & Martin, T. A. (2015). Architectures of adaptive integration in large collaborative projects. *Ecology And Society*, 20(4).
<https://doi.org/10.5751/es-07788-200405>

References:

Klein, J. T. (2018). Learning in transdisciplinary collaborations: A conceptual vocabulary. In: Fam, D., Neuhauser, L., & Gibbs, P. (red.). Transdisciplinary theory, practice and education: The art of collaborative research and collective learning, pp. 11-23. Springer.

OECD (2021), Embedding Values and Attitudes in Curriculum: Shaping a Better Future, OECD Publishing, Paris, <https://doi.org/10.1787/ae2adcd-en>

Practice

Currently, several examples can already be found in the Netherlands. Both universities of applied sciences and research universities offer transdisciplinary courses. A few examples of transdisciplinary education:

- In the minor [Act for Change Together](#) at Avans University of Applied Sciences, students from various disciplines collaborate with stakeholders on societal issues related to the new economy.
- At NHL Stenden University of Applied Sciences, students can participate in the [Social Innovation Lab](#) project, located in the heart of a neighborhood, where they work directly with residents to apply their practice collaboratively.
- In the master's insert [Shaping Responsible Futures](#) at the University of Twente, students work together with municipalities, NGOs, or companies on issues concerning responsible futures and systemic change.
- In the elective course [Placemaking](#) at the University of Amsterdam, students from different backgrounds engage with local residents and stakeholders to tackle urban challenges and co-create solutions.